GROUP 13

Smart Travel & Camping Guider

Introduction & Background

- Outdoor traveling and camping are becoming increasingly popular these days.
- Research also shows that around 41% of hikers become lost during outdoor activities.

  Japan reports over 1,000 missing outdoor travelers each year

- There is a need for a smart travel tracking system for travelers .

Problem Statement

During hiking and outdoor activities, travelers can easily get separated from their group and take long time to find them due to several environmental and communication-related issues.

BAD WEATHER

Rain, fog, and storms reduce visibility and cause separation.

WEAK VISIBILITY

Low light and mist make it hard to see group members.

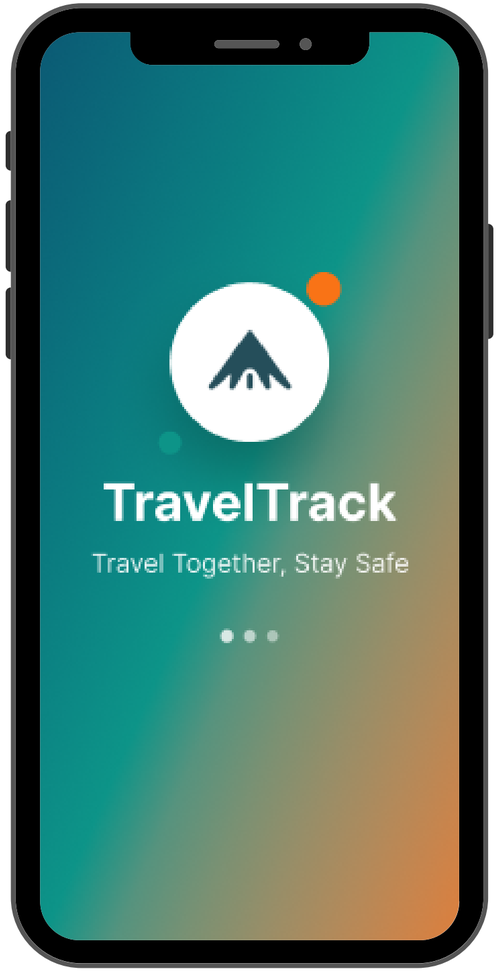
COMMUNICATION PROBLEMS

Weak network and device issues affect communication.

FOREST ENVIRONMENTS

Dense forests and unclear paths lead to getting lost.

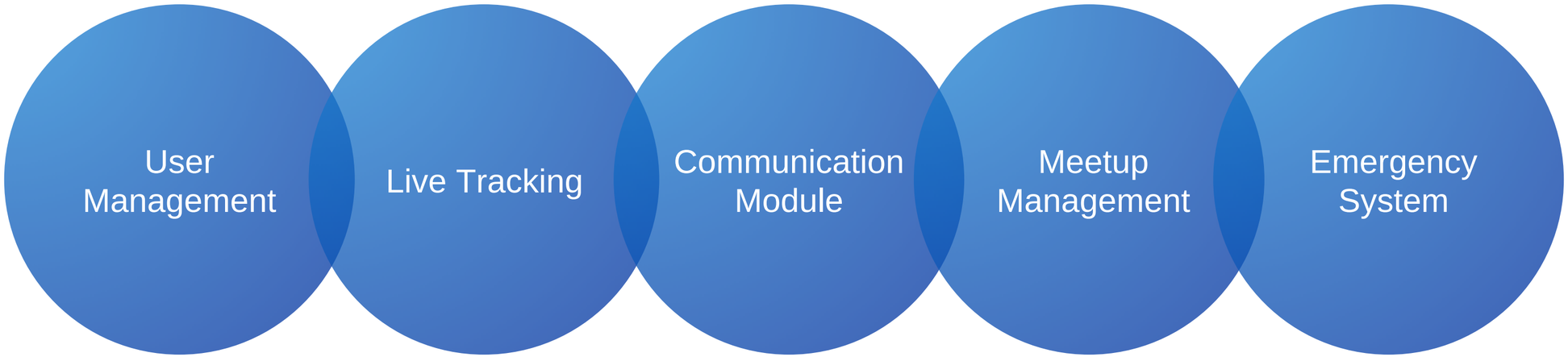
Our Proposed Solution



The System improves traveler safety and group coordination using real-time tracking and communication during outdoor activities.

- 50-METER SAFE RADIUS MONITORING
- LOST TRAVELER DETECTION
- MEETUP POINT SHARING
- GROUP MESSAGING SYSTEM
- SOS EMERGENCY ALERT SYSTEM
- OFFLINE-FRIENDLY COMMUNICATION SUPPORT

Project Scope

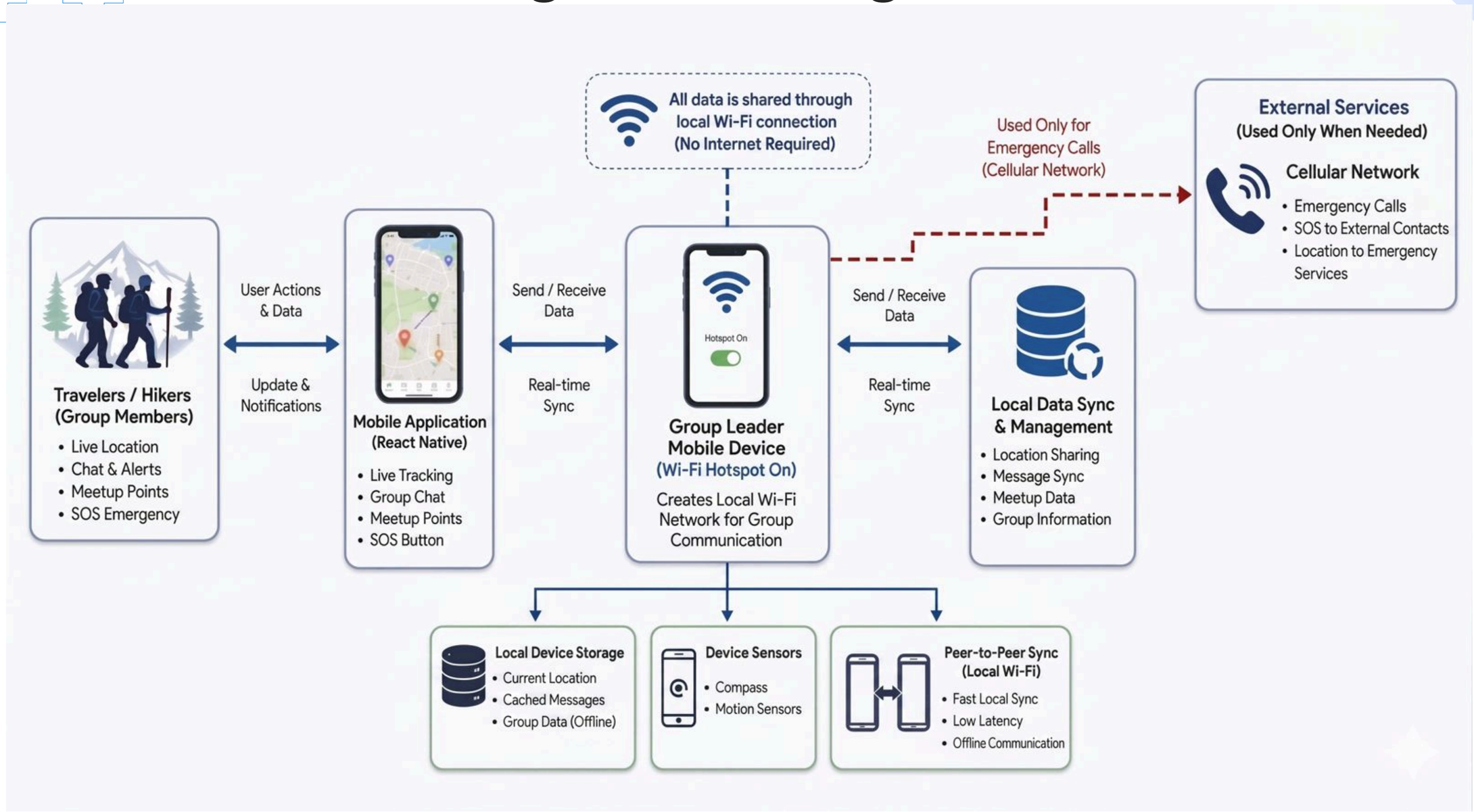


Objectives

Develop a smart mobile application to improve traveler safety and coordination during hiking and camping activities.

- **Detect users leaving a 50m safe radius**
- **Enable meetup point sharing**
- **Support real-time group communication**
- **Provide one-touch SOS emergency alerts**

High Level Diagram



Technology Stack

Frontend

React Native / Flutter

Backend

Firebase Authentication, Firebase Firestore, Firebase Cloud Messaging

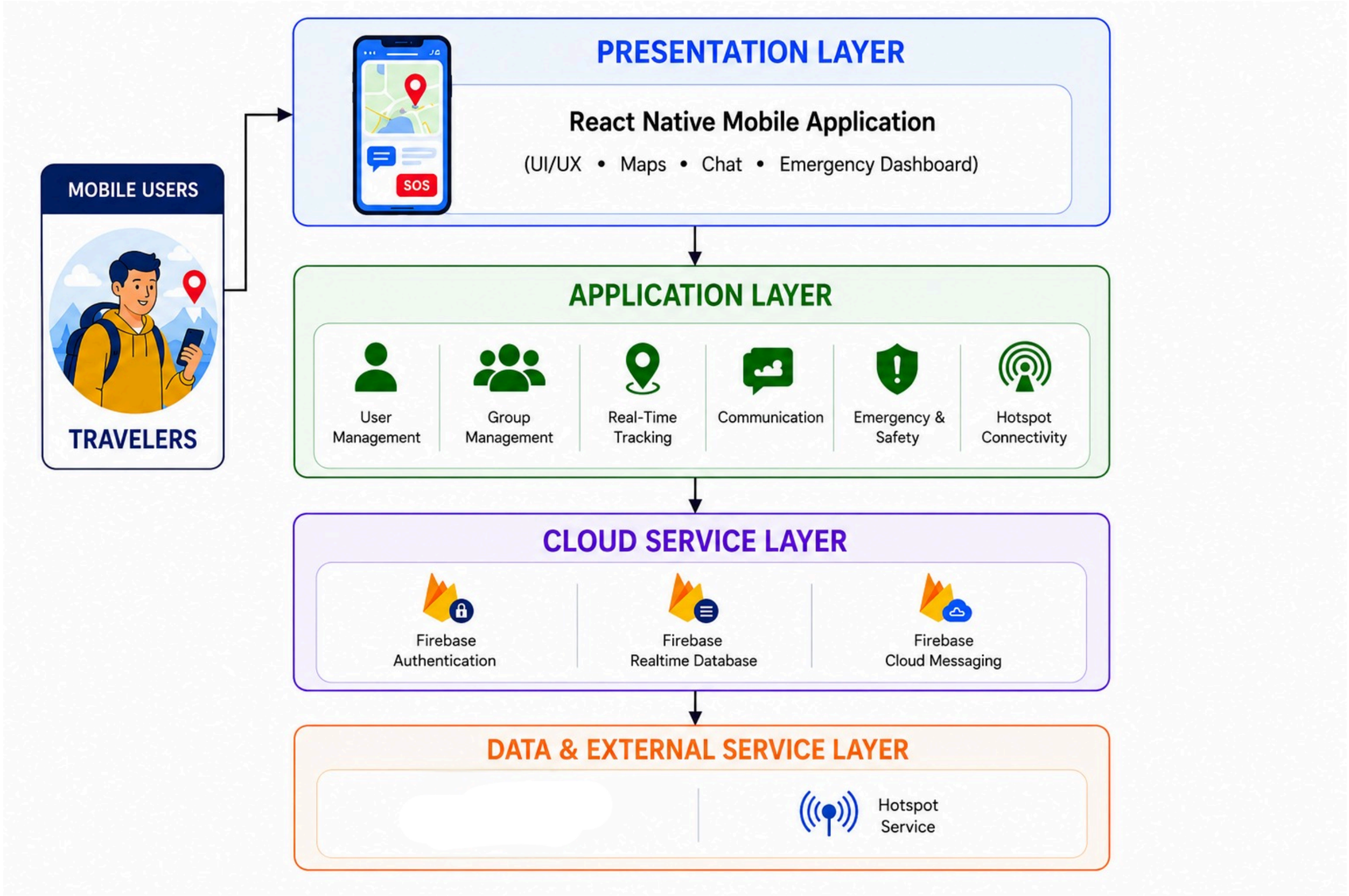
APIs & Services

Wi-Fi RTT (Round Trip Time)

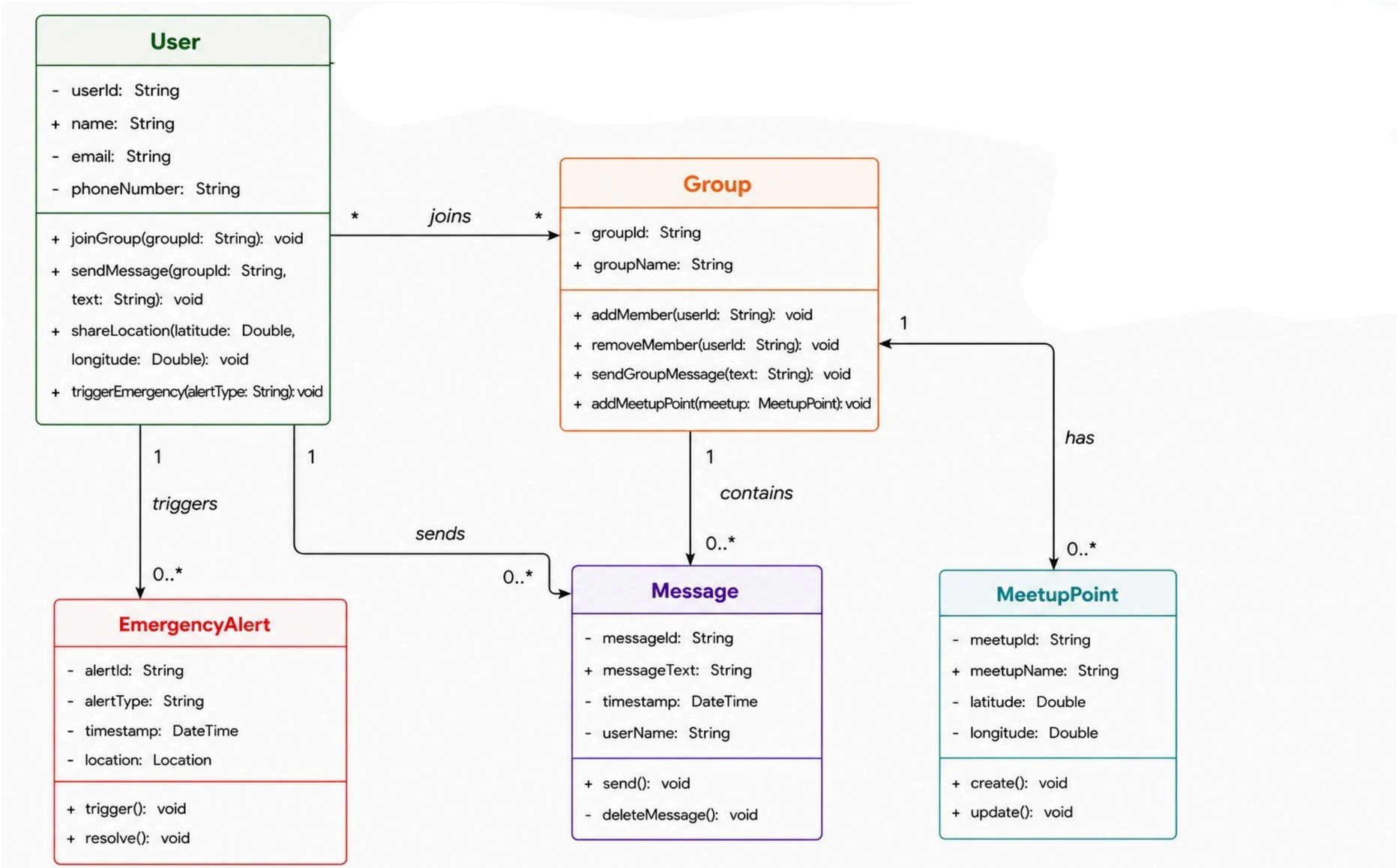
Development Tools

Visual Studio Code, GitHub, Figma

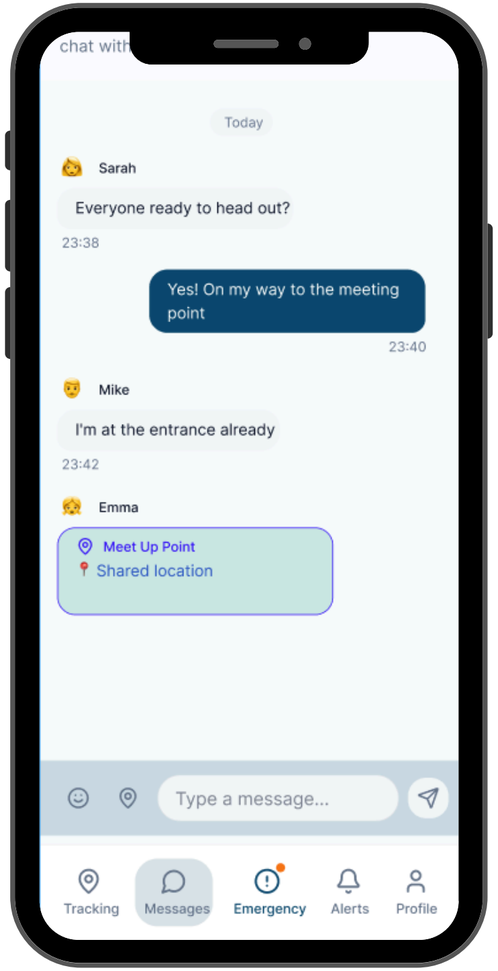
System Architecture

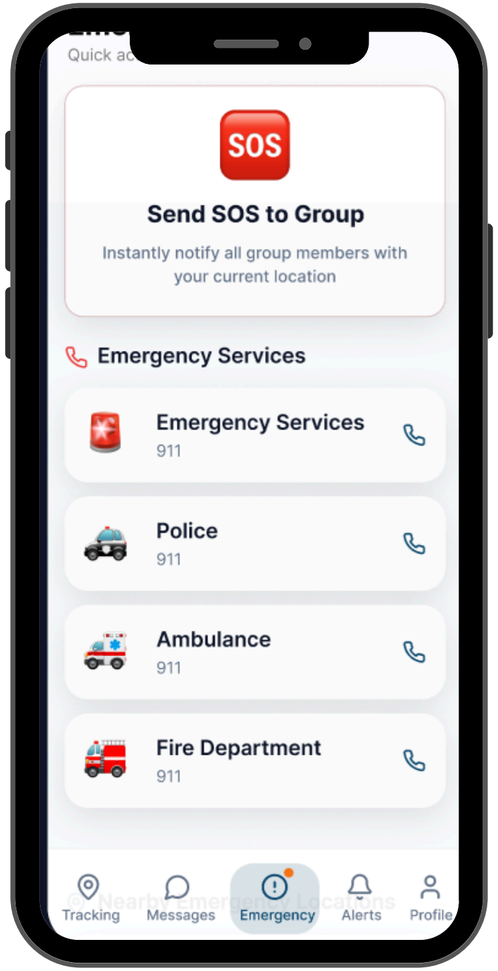


Class Diagram



Times Square
Active Friends
3 online
Sarah
Mike
Emma
Jack
You
Jack
Mike
Sarah
Emma
Send Meetup
SOS Alert
Tracking
Messages
Emergency
Alerts
Profile


Today
Sarah
Everyone ready to head out?
23:38
Yes! On my way to the meeting point
23:40
Mike
I'm at the entrance already
23:42
Emma
Meet Up Point
Shared location
Type a message...
Tracking
Messages
Emergency
Alerts
Profile


SOS
Send SOS to Group
Instantly notify all group members with your current location
Emergency Services
Emergency Services
911
Police
911
Ambulance
911
Fire Department
911
Tracking
Messages
Emergency
Alerts
Profile

Expected Deliverables

- Fully functional mobile application with
  - Real-time traveler tracking
  - Group communication module
  - SOS emergency alert system
- UI/UX design prototypes
- System documentation
- Final presentation

Project Timeline

Week 1
Requirement Analysis

Week 2–3
System Setup

Week 4–5
Backend Development

Week 6–7
Tracking System

Week 8–9
Communicatio n Module

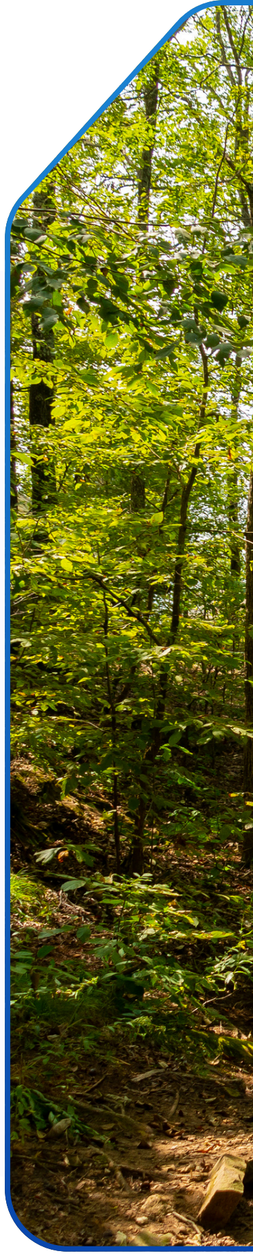
Week 10
Emergency System

Week 11
Testing

Week 12
Optimization

Week 13
Final Deployment

1 2 3 4 5 6 7 8 9



Thank You